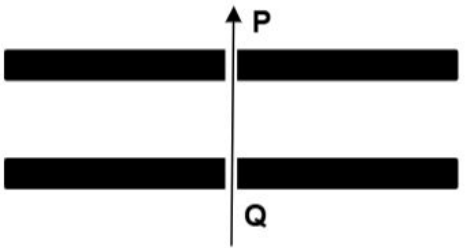
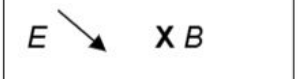
25 An electron moving in a straight line passes through small holes P and Q in two identical plates.



There is a uniform electric field *E* and a uniform magnetic field *B* between the two plates.
The directions of these fields are shown by arrows or these symbols:
⊙ a field coming out of the paper; **X** a field going into the paper.

Which orientation of the two fields is **not** possible?

	A		B		C		D	

L2 **Answer: B**

Option A: Possible. No effect of B field since it is parallel to motion of electron. Electron can move in straight line accelerated by E field

Option B: Not possible. Magnetic force on electron to the left. Due to electric field, electrostatic force on electron will be out of the plane of paper. There is a net force on the electron that changes as it moves which makes it not possible for it to go in a straight line out of P.

Option C: Possible. Magnetic force on electron to the right due to B field. Due to E field, there is a component of force on electron to the left (horizontal component) and upward (vertical component). It is possible for electron to move straight if the horizontal component of electric force (leftward) balances the magnetic force (rightward).

Option D: Possible. There is electric force on electron into the plane of paper. Due to magnetic field, there is magnetic force out of the paper. If these two forces are equal in magnitude, the net force on the electron will be zero and it will continue to move in a straight line.